



CENG 408

Innovative System Design and Development II

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**MULTI-LABEL CHEST X-RAY DISEASE CLASSIFICATION
AND EXPLAINABILITY WITH DEEP LEARNING**

DISCUSSION

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Discussion

1. Interpretation and Comparison of Results

The experimental results show a clear evolution in diagnostic performance through architectural improvements. The transition from standard BCE loss to a customized pipeline in Model 4 demonstrates how challenges like class imbalance can be addressed. While Model 1 achieved high accuracy, it suffered from the Accuracy Paradox, where the network stayed accurate simply by predicting negative for most cases. The implementation of Asymmetric Loss (ASL) in Model 2 was a major turning point, forcing the network to focus on rare pathologies by penalizing missed diseases. Although Model 3 attempted to use preprocessing like CLAHE and Gamma Correction, performance actually dropped, suggesting that the CBAM attention mechanism is more reliable on raw images. Finally, Model 4 reached the best performance by replacing the rigid 0.5 decision boundary with Dynamic Class-Specific Thresholds. This proves that optimizing thresholds based on the visual characteristics of each disease is the ultimate factor in unlocking the model's full diagnostic potential.

1.1. The Failure of Standard BCE (Model 1)

Model 1 used the standard BCEWithLogitsLoss, which resulted in the "Accuracy Paradox." Although the average accuracy was high (95.22%), the Macro F1 score was very low (0.2736). This indicates that the model was simply predicting "Healthy" for most cases to minimize loss, effectively ignoring rare diseases like Pneumothorax.

1.2. The Impact of Asymmetric Loss (Model 2)

The implementation of Asymmetric Loss (ASL) in Model 2 was a turning point. By penalizing the network more for missing positive cases and reducing the weight of easy negative samples, the Macro F1 score improved significantly to 0.3458. This shift proved that the model began focusing on actual pathological features rather than just background pixels.

1.3. Preprocessing Limitations (Model 3)

Contrary to our initial hypothesis, applying CLAHE and Gamma Correction in Model 3 did not improve results; instead, the Macro F1 score dropped to 0.3307. The CBAM (Attention Module) likely focused on artificial noise artifacts created by the enhancement, rather than the real diseases. This confirms that raw image quality is more reliable for this specific deep learning architecture.

1.4. The Success of Dynamic Thresholds (Model 4)

Model 4 achieved the best performance with a Macro F1 score of 0.3592. The main success here was replacing the rigid 0.5 decision threshold with class-specific thresholds. For example, lowering the threshold to 0.35 for Pneumonia allowed the model to detect faint signals it previously missed. This model approaches the level of human radiologist agreement, making it a robust tool for clinical support.

2. Limitations

While the final model achieved success, several limitations affected the development process. These constraints involve both the computational environment used for training and the characteristics of the medical dataset.

- **Hardware and Environment Constraints:** The model was trained using Google Colab, which imposed strict daily GPU usage limits and restricted the total training time. Frequent network instability and session timeouts made it difficult to maintain long, uninterrupted training cycles.
- **Memory and Batch Size Restrictions:** Due to system memory limits, the training was restricted to a small batch size of 8. This affected the stability of gradient updates compared to larger-scale hardware setups.
- **Image Resolution:** Computational limits required input images to be downscaled from 1024x1024 to 512x512 pixels. This reduction in resolution might cause the model to miss extremely small pathological features, such as tiny nodules.
- **Complexity of Specific Diseases:** Certain pathologies, like Pneumonia, remain difficult to classify accurately because their visual signatures often overlap with other lung patterns. Even with dynamic thresholds, these diseases show lower F1 scores compared to visually distinct conditions like Cardiomegaly.

3. Ethical Concerns:

The integration of this system into a mobile application brings ethical responsibilities regarding data privacy and the role of AI in healthcare. This system works as a decision-support tool and does not replace medical professionals. The final diagnostic decision always belongs to the radiologist. To protect patient privacy, the application requires secure login for authorized doctors and uses anonymous images to follow healthcare standards. Although the model shows high performance, it is important to check for biases in the training data by testing it on different populations. Finally, the system provides probability scores and dynamic thresholds to help doctors understand the predictions. This creates a transparent and accountable workflow between the technology and the doctor.

4. Scalability

The system is designed to handle more data and more users in the future. We use MongoDB as the primary database to store medical records and model results. This NoSQL structure makes it easy to add new data fields without slowing down the system. As the number of users increases, the database can scale to maintain performance. Also, the modular design of the application allows us to add new models to detect more diseases in the future. The network infrastructure ensures that the mobile application can support many doctors across different locations at the same time.